Cross-Scene Hyperspectral Image Classification Network With Dynamic Perturbation and Self-Knowledge Distillation

Yuhang Hong, Zhixi Feng, *Member, IEEE*, Shuyuan Yang, *Senior Member, IEEE*, and Zhihao Chang

***Abstract*—Cross-scene hyperspectral image (HSI) classification faces spectral–spatial feature distribution shifts resulting from cross-domain heterogeneity, which has become a critical challenge that urgently needs resolution in the field of remote sensing intelligent interpretation. To address this distribution shift, the mainstream approach is domain generalization (DG). However, existing HSI DG methods primarily focus on interclass separability, while paying relatively less attention to cross-domain transferability. To overcome the limitation that existing methods mainly focus on interclass separability, this study proposes a cross-scene HSI classification network, termed DPSKDnet. By synergistically employing dynamic perturbation-based destylization and self-knowledge distillation modeling mechanisms, DPSKDnet constructs domain-invariant representations with strong generalization capabilities in the feature space. Specifically, this study first builds a generator based on dynamic perturbation destylization to mine source domain (SD)-invariant features and generate extended domain (ED) samples. Subsequently, a Fourier augmentation module (FAM) is utilized to optimize the frequency-domain representations of the SD, ED, and their combination-generated intermediate domain (ID), obtaining frequency-enhanced representations. To effectively improve the model's ability to capture domain-invariant features, a sample pair distillation loss is devised. This loss, informed by multidomain mixed data input, guides the discriminator in online self-supervised learning. The overall accuracy (OA) of this method on Loukia, Houston2018, and Pavia Center (PC) increased by 0.48%, 0.81%, and 1.5%, respectively, compared to state-of-the-art methods. The code is available on the website https://github.com/Yuhang-Hong/TGRS_DPSKDnet**

***Index Terms*—Cross-scene hyperspectral image (HSI) classification, domain generalization (DG), dynamic perturbation, generative adversarial network, self-knowledge distillation.**

I. Introduction

HYPERSPECTRAL images (HSIs), characterized by their rich spectral information, have been widely applied in remote sensing, agriculture, and environmental monitoring [1]. The rapid development of deep learning techniques has substantially propelled progress in HSI classification in recent years, leading to the demonstration of remarkable performance [2], [3]. However, most existing methods heavily rely on extensive labeled data and operate under the assumption that the training and testing data adhere to an independent and identically distributed (i.i.d.) assumption [4], [5]. Nonetheless, this assumption is often not upheld in real-world applications, thereby constraining the generalizability of these models. First, acquiring high-quality labeled samples requires substantial human effort and is often cost-prohibitive. Second, due to variations in sensor conditions, acquisition environments, and geographic regions, even samples from the same class may exhibit significant differences in spectral reflectance. These factors lead to a distribution discrepancy between the training set [source domain (SD)] and the testing set [target domain (TD)], a phenomenon commonly referred to as domain shift. When a model trained on the SD is directly applied to the TD, this domain shift can severely degrade its performance.

Currently, the most widely adopted method for addressing domain shift is domain generalization (DG). DG methods are typically formulated within a feature mapping framework, where the model is trained by jointly optimizing empirical risk minimization and a regularization term that promotes generalization [6]. Acquiring multiple hyperspectral SD datasets is also challenging in practice. As a result, single-source DG has emerged as a more promising and applicable approach in cross-scene HSI classification. Recent studies on single-source HSI DG [7], [8], [9], [10] have made progress, primarily focusing on constructing synthetic domains, incorporating external semantics, or optimizing feature alignment. However, most of these methods primarily focus on enhancing interclass separability (e.g., through contrastive learning or semantic integration), while paying comparatively less attention to cross-domain transferability. This transferability is crucial for enhancing the generalization ability of the model to unseen scenarios, as it ensures consistent and discriminative representations across diverse hyperspectral domains, thus enabling more reliable cross-domain classification.

To address the limitation that existing methods mainly focus on interclass separability, this study proposes a cross-scene HSI classification network, termed DPSKDnet. The proposed network integrates dynamic perturbation and self-knowledge distillation to maintain interclass separability while enhancing cross-domain transferability. Specifically, to improve cross-domain adaptability, DPSKDnet introduces a generator based on dynamic perturbation destylization that applies dynamic

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: zxfeng@xidian.edu.cn).

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noise to the original samples and removes domain-specific information through a destylization strategy. This process produces extended domain (ED) samples that preserve semantic consistency while exhibiting higher diversity, making them closer to domain-invariant representations. Based on this, the network fuses SD samples with ED samples to construct intermediate domain (ID) samples. A Fourier augmentation module (FAM) is then employed to optimize the frequency-domain representations of SD, ED, and ID samples, thereby enhancing feature robustness and cross-DG. Furthermore, to establish domain-invariant knowledge across different domains, a sample pair distillation loss is proposed to guide the discriminator in performing cross-domain online self-supervised learning. Specifically, samples from different domains but with the same labels are paired, and their mean logits serve as a cross-domain consensus. The model penalizes predictions that deviate from this consensus, preventing bias toward any specific domain and encouraging the learning of domain-invariant representations. In addition, to preserve interclass separability, a supervised contrastive loss is incorporated into the discriminator, encouraging intraclass compactness and interclass dispersion. This design enables DPSKDnet to achieve a balanced improvement between cross-domain adaptability and discriminative capability.

The primary contributions of this study are as follows.

1) This study proposes a generator based on dynamic perturbation and destylization for modeling SD-invariant features. Specifically, a learnable dynamic mixing strategy is first adopted to enhance the diversity of feature representations. Subsequently, Gaussian noise is added to expand the perturbation range, which is used to simulate variations within the latent space. Finally, by employing global channel mean and variance to normalize the perturbed features, style-related information is removed, ultimately yielding a domain-invariant feature representation.
2) To enrich the representation of the original data, we employ the 2-D discrete Fourier transform to map each spectral channel of the original data into the frequency domain. Within the amplitude spectrum and the phase spectrum, we introduce multiplicative and additive noise to construct spectrally enhanced representations, thereby forming a multidomain sample set.
3) To construct domain-invariant knowledge across diverse domains, a novel sample pair distillation loss, termed $\mathcal{L}_{\text{pairKD}}$, is proposed. Specifically, $\mathcal{L}_{\text{pairKD}}$ operates by pairing data samples that share the same label but originate from different domains. The average logits of the paired domains are then utilized as guidance to penalize samples that are inconsistent with this consensus knowledge. This measure effectively prevents the learning process from becoming biased toward any specific domain, thereby strongly encouraging the acquisition of domain-invariant features.

The remainder of this study is organized as follows. Section II introduces the relevant work on DG for HSIs and concepts related to knowledge distillation. Section III elaborates on the proposed DPSKDnet method. Section IV presents the experimental results and analysis, followed by the conclusions drawn in Section V.

II. Related Work

A. DG in HSI Cross-Scene Classification

To address the domain shift, practical HSI classification tasks often adopt a cross-scene classification strategy, in which a model trained on labeled source scenes is used to predict unlabeled target scenes. However, traditional methods exhibit limited effectiveness under cross-scene settings. The fundamental reason lies in their reliance on the i.i.d. assumption during training, which hinders their ability to generalize effectively to out-of-distribution target scenes. Current research on cross-scene classification primarily focuses on two approaches [11]: Domain adaptation (DA) [12] and DG [13]. DA techniques aim to transfer models from the SD to the TD by aligning the data distributions of the two domains [14], [15]. Although this approach can alleviate domain shift to some extent, its dependence on TD data limits its practical applicability. On the one hand, TD data may be difficult or costly to obtain; on the other hand, the need to retrain the model whenever a new scene is encountered conflicts with the demand for time-efficient deployment in real-world scenarios. Given that TD data are often unavailable in real-world applications, cross-scene HSI classification tasks increasingly favor DG approaches [16]. Compared to DA methods that rely on access to TD data, DG is more challenging but also more aligned with practical requirements. The core idea of DG is to train models using data from one or more SDs only, enabling the learning of domain-invariant feature representations that generalize effectively to entirely unseen TD [13].

Constrained by the high cost and inherent complexity of hyperspectral data acquisition (encompassing factors such as sensor calibration and environmental interference), the construction of multi-SD HSI datasets poses significant challenges. This technical bottleneck directly contributes to the current focus of DG algorithms in cross-scene HSI classification research, primarily exploring the single-SD paradigm. Zhang et al. [7] were the first to introduce a single-source DG algorithm to cross-scene HSI classification, aiming to mitigate overfitting from a data manipulation perspective to enhance the model's generalization performance. Subsequently, Zhao et al. [10] approached the problem from the perspective of enhancing domain alignment algorithms, proposing LLURnet to improve the robustness of domain-invariant feature learning. Qin et al. [17] focused on the data geometry aspect, constraining the semantic information degradation caused by excessive randomization and proposed FDGnet. Furthermore, several works [8], [18] have adopted a multimodal perspective, integrating language-aware semantic prototypes into visual embeddings to guide the learning of domain-invariant features. Dong et al. [19] addressed the spectral shift issue by designing a generator to simulate spectral and spatial biases, employing a causal alignment approach to achieve efficient DG. Cui et al. [20] adopted a lightweight perspective, designing an attention-assisted DG network to improve model performance while reducing resource consumption. Currently, most research

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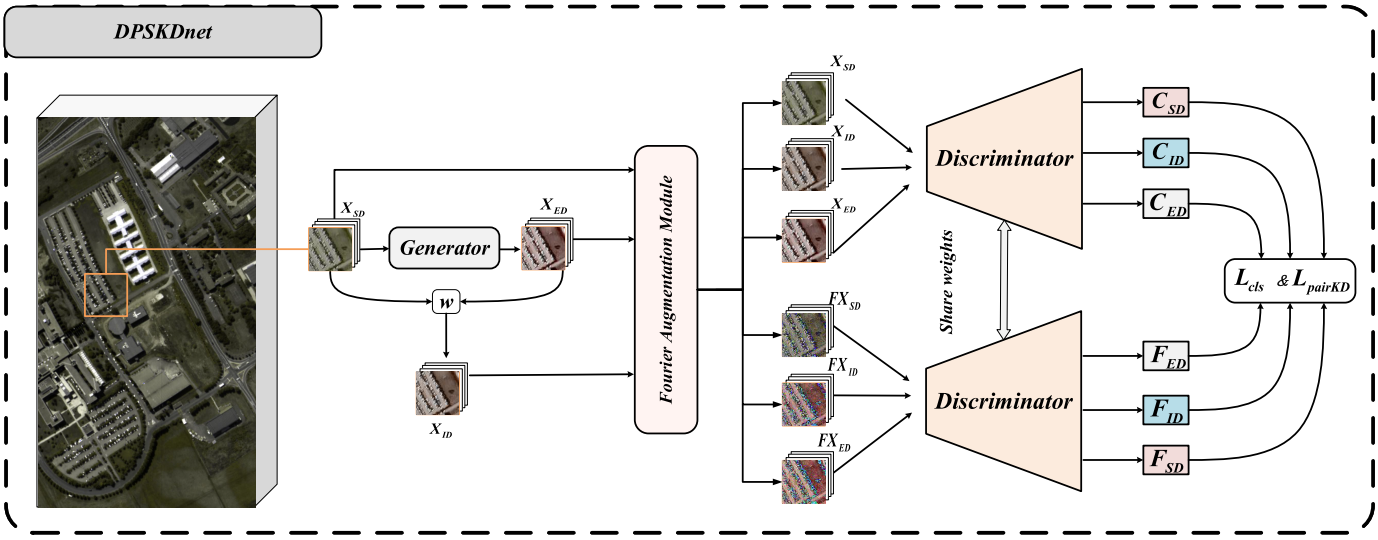


Fig. 1. Flowchart of the proposed DPSKDnet. The DPSKDnet network adopts a generative adversarial network framework, with its architecture primarily comprising three components: 1) a generator based on dynamic perturbation destylization, responsible for mapping SD data to the ED space; 2) an FAM, which simultaneously outputs the original data and its frequency-enhanced representation by performing a frequency-domain transformation and feature reinforcement on the input data; and 3) a discriminator integrated with a self-knowledge distillation mechanism, which outputs classification vectors for the input multidomain data.

focuses on convolutional neural networks, but some studies are also concentrating on the Mamba [21] architecture [22], [23], [24]. Huang et al. [25] proposed DTAM, aiming to capture global image features for superior generalization performance.

B. Knowledge Distillation

Knowledge distillation, introduced by Hinton in 2015 [26], is a technique whose fundamental principle involves transferring the "knowledge" from a pretrained, complex teacher model to a more lightweight student model. Knowledge distillation leverages the soft labels generated by the teacher model—specifically, the probability distribution over classes—to convey richer information, including interclass similarities. Given an input data point x_i and its corresponding label $y \in \{1, \ldots, K\}$, the logits output of the model is represented as $z(x) = [z_1(x), \ldots, z_K(x)]$. The posterior predictive distribution for the data x can be expressed as

$$P(y \mid x; \theta, \tau) = \frac{\exp\left(z_y(x)/\tau\right)}{\sum_{i=1}^{K} \exp\left(z_i(x)/\tau\right)} \tag{1}$$

where the model is parameterized by θ and τ is the temperature scaling factor.

For traditional knowledge distillation, this is achieved by minimizing the Kullback–Leibler (KL) divergence between the predicted distributions, as shown in the following formula:

$$\mathcal{L}_{\mathrm{KD}}(X; \theta_s, \tau) = \sum_{x_i \in X} \sum_{k=1}^{K} D_{\mathrm{KL}}\left(P(k \mid x_i; \theta_t, \tau) \| P(k \mid x_i; \theta_s, \tau)\right) \tag{2}$$

where X represents a batch of input data and θ_t and θ_s denote the parameters of the teacher and student models, respectively.

Diverging from conventional knowledge distillation, self-knowledge distillation is founded on the principle of internal knowledge extraction from the model itself, consequently eliminating the requirement for an external teacher model [27]. Self-knowledge distillation serves as an effective paradigm for both model compression and performance enhancement, with its typical implementation methods primarily falling into three categories: self-knowledge distillation based on intermediate features [28], [29]; self-knowledge distillation based on self-attention mechanisms [30]; and self-knowledge distillation based on prediction consistency [31], [32]. This study primarily focuses on self-knowledge distillation based on prediction consistency, the detailed exposition of which is provided in Section III.

III. Methodology

This section details the proposed DPSKDnet for cross-scene HSI classification. For the cross-scene HSI classification task, the input SD sample set can be represented as $X_{\mathrm{SD}} = \{x_i\}_{i=1}^{N} \in \mathbb{R}^{N \times d \times p \times p}$, where $x_i \in \mathbb{R}^{d \times p \times p}$ comprises $p \times p$ spatial dimensions and d spectral features. The corresponding class labels for X_{SD} can be denoted as $Y_{\mathrm{SD}} = \{y_i\}_{i=1}^{N}$, where N represents the batch size of samples input to the network. The proposed DPSKDnet consists of three primary components: a generator, an FAM, and a discriminator, as illustrated in Fig. 1. Specifically, the SD sample set X_{SD} is processed by the generator based on dynamic perturbation destylization to yield the ED sample set X_{ED}. The ED sample set X_{ED} is linearly combined with the SD sample set X_{SD} to generate the ID sample set X_{ID}. The three domain sample sets, X_{SD}, X_{ID}, and X_{ED}, are then fed into the FAM, which retains the original data X_{SD}, X_{ID}, and X_{ED} while also generating frequency-enhanced representations $\mathrm{FX}_{\mathrm{SD}}$, $\mathrm{FX}_{\mathrm{ID}}$, and $\mathrm{FX}_{\mathrm{ED}}$, collectively forming the multidomain sample set. The discriminator, integrated with a self-knowledge distillation mechanism, produces a classification tensor $C \in \mathbb{R}^{N \times K}$ and $F \in \mathbb{R}^{N \times K}$ for the multidomain input, where N is the batch size and K is the number of classes.

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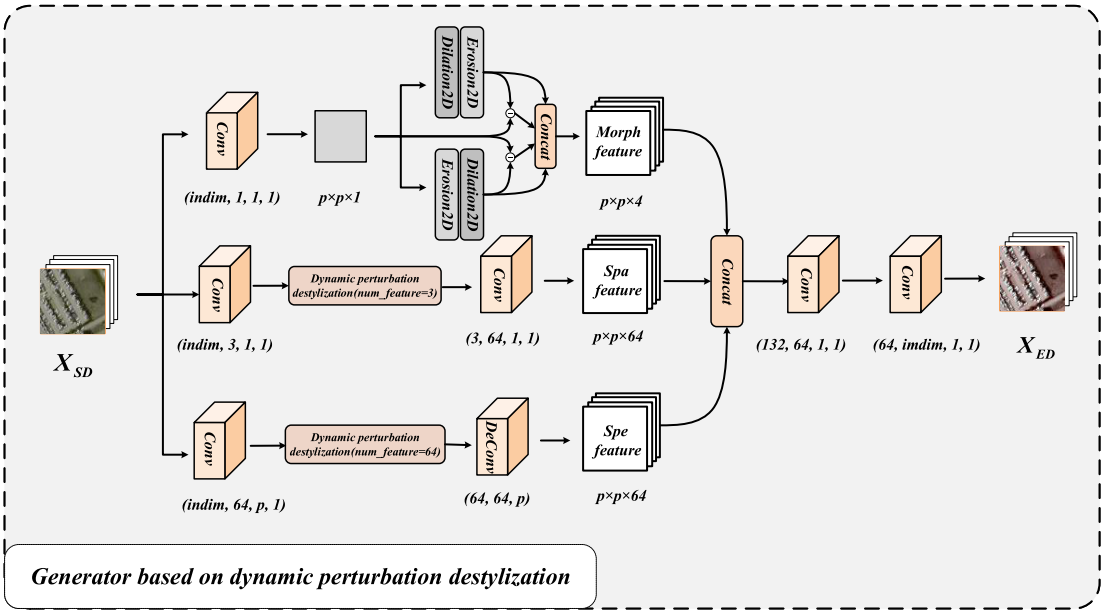


Fig. 2. Structure of the generator is based on dynamic perturbation destylization. The generator comprises three parallel branches, each dedicated to extracting morphological, spatial, and spectral features, respectively. The extracted features are subsequently concatenated and processed via convolutional operations to reconstruct the ED samples.

The classification tensor C and F are utilized to compute the classification loss $\mathcal{L}_{\text{cls}}$ and the sample pair distillation loss $\mathcal{L}_{\text{pairKD}}$.

A. Generator Based on Dynamic Perturbation Destylization

In DG scenarios, the generator is designed to produce augmented samples with domain-invariant features, which effectively bridge the distribution gaps between different domains in the feature space. To mitigate the model's over-reliance on domain-specific local features, we propose a generator based on dynamic perturbation destylization. By applying dynamic noise to SD samples and incorporating a destylization mechanism, the generator produces ED samples that exhibit higher diversity and stronger domain invariance, thereby encouraging the model to focus on global semantic features. The architecture of the generator is illustrated in Fig. 2. Taking the SD sample set X_{SD} as input, the generator employs parallel feature extraction pathways to learn feature representations across different dimensions, including morphological, spatial, and spectral features. These features are then concatenated and refined via convolutional operations to reconstruct the ED sample set X_{ED}. By extracting and integrating three complementary types of features from the SD, the generator effectively constructs domain-invariant representations, serving as a critical component in improving generalization under domain shift conditions.

Morphological features primarily capture global semantic information, also known as template features. In this study, basic morphological operations, dilation and erosion, are employed for morphological feature extraction. Specifically, before feature extraction, the dimensionality of the input SD samples is first transformed into a grayscale image format of $p \times p \times 1$ via a convolutional operation. This transformation is based on the findings of Mondal et al. [33], which indicate the significant processing efficacy of morphological operators on grayscale images. This morphological feature extraction branch comprises two core modules: Dilation2D and Erosion2D. For the dilation operation, given an input tensor $I \in \mathbb{R}^{B\times C\times H\times W}$ and a kernel $M \in \mathbb{R}^{3\times 3}$, the dilation operation is defined as

$$\textbf{Dilation}\,(I, M)\,(p, q) = \max_{(i,j)\in M} I\,(p + i, q + j) \tag{3}$$

the dilation operation takes the maximum value within the kernel coverage area. For the erosion operation, it can be defined as

$$\textbf{Erosion}\,(I, M)\,(p, q) = \min_{(i,j)\in M} I\,(p + i, q + j) \tag{4}$$

the erosion operation takes the minimum value within the kernel coverage area. By integrating the output features from both the Dilation2D and Erosion2D morphological operations and concatenating them, a comprehensive morphological feature representation is ultimately constructed.

The processing of spatial and spectral features is designed to obtain style-independent representations, that is, to achieve destylization. In the spatial feature extraction stage, a 1×1 convolutional layer is first applied to reduce the dimensionality of spatial patches, compressing them into a $p \times p \times 3$ feature map, which serves as a compact representation of spatial information. This feature map is then processed by the dynamic perturbation destylization module and subsequently reconstructed into the final spatial feature representation via convolutional layers. Specifically, the input feature map is first subjected to a learnable dynamic mixing strategy, which enhances the diversity of the feature expressions. Next, Gaussian noise sampled from a normal distribution is added to the mixed features, further expanding the perturbation range to simulate variations across potential latent domains.

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The channel-wise mean and variance of the perturbed feature map are then computed, and global statistics are updated using a momentum-based strategy. Finally, inspired by destylization techniques [34], the perturbed features are normalized using the global channel mean and variance, effectively removing style-related information while preserving structural content. This process yields a spatial feature representation with improved domain invariance.

This method primarily encompasses four steps: dynamic mixing mechanism, dynamic perturbation injection, dynamic statistics updating, and normalization. Given an input tensor $X \in \mathbb{R}^{N\times C\times H\times W}$, where N denotes the batch size, and C, H, W correspond to the number of channels and spatial dimensions, respectively. The dynamic mixing mechanism generates new, diverse samples by mixing the original input features with randomly permuted features. The dynamic mixing mechanism can be formally represented as

$$X_{\text{mix}} = \varphi X + (1-\varphi)\, X_{\sigma} \tag{5}$$

where φ is a learnable mixing parameter initialized to 0.5. X_{σ} represents the randomly permuted input tensor. After dynamic mixing, the tensor X_{mix} then undergoes dynamic perturbation. The core purpose of dynamic perturbation is to inject learnable randomness while preserving the original content information, thereby helping the model achieve domain invariance learning. The formula for dynamic perturbation is as follows:

$$X_{\text{perturbed}} = X_{\text{mix}} + \gamma\Delta \tag{6}$$

where $\Delta \sim \mathcal{N}(0, I)$ represents standard Gaussian noise. γ is a learnable perturbation intensity parameter initialized to 0.1. $X_{\text{perturbed}}$ is used to calculate the channel-wise mean and channel-wise variance for momentum updating. The formula for the channel-wise mean is as follows:

$$\mu_b^{(c)} = \frac{1}{\text{NHW}} \sum_{n,h,w} X_{\text{perturbed}}^{(n,c,h,w)} \tag{7}$$

and the formula for channel-wise variance is as follows:

$$\sigma_b^{2(c)} = \frac{1}{\text{NHW}} \sum_{n,h,w} \left(X_{\text{perturbed}}^{(n,c,h,w)} - \mu_b^{(c)} \right)^2 \tag{8}$$

after obtaining the channel-wise mean and channel-wise variance, momentum updating is employed to update the global channel-wise mean and global channel-wise variance. The formula for momentum updating is as follows:

$$\begin{aligned} \mu_{\text{global}}^{(c)} &\leftarrow \eta \cdot \mu_{\text{global}}^{(c)} + (1-\eta)\cdot \mu_b^{(c)} \\ \sigma_{\text{global}}^{2(c)} &\leftarrow \eta \cdot \sigma_{\text{global}}^{2(c)} + (1-\eta)\cdot \sigma_b^{2(c)} \end{aligned} \tag{9}$$

where η is the momentum parameter, the momentum-updated global channel-wise mean and global channel-wise variance are then used for normalization, with the normalization formula as follows:

$$X_{\text{out}}^{(n,c,h,w)} = \frac{X_{\text{perturbed}}^{(n,c,h,w)} - \mu_{\text{global}}^{(c)}}{\sqrt{\sigma_{\text{global}}^{2(c)} + \epsilon}} \tag{10}$$

where $X_{\text{out}}^{(n,c,h,w)}$ represents the value of the perturbed tensor at batch index n, channel c, and spatial location (h, w). $X_{\text{perturbed}}^{(n,c,h,w)}$ represents the value of the perturbed input tensor at the corresponding position. $\mu_{\text{global}}^{(c)}$ represents the global mean of channel c, accumulated across historical batches using the momentum method. $\sigma_{\text{global}}^{2(c)}$ represents the global variance of channel c, similarly accumulated using the momentum method. ϵ represents a numerical stability term, included to prevent division-by-zero errors. By using global, stable statistics for normalization, the model can map features with varying local styles (i.e., different $X_{\text{perturbed}}$) onto a standardized distribution, thereby effectively removing style discrepancies and promoting the learning of domain-invariant structural features.

In the spatial feature extraction section, the feature dimension C is set to 3 as the channel parameter for dynamic perturbation destylization. In the spectral feature extraction section, leveraging the channel-wise information aggregation property, a 1×1 convolutional layer is first employed to compress the features to a $1 \times 1 \times 64$-D. Subsequently, a dynamic perturbation destylization operation with a channel parameter of 64 is performed, and finally, a deconvolutional layer is used to restore the spatial dimensions to $p \times p \times 64$. Finally, the morphological features, spatial features, and spectral features are concatenated along their feature dimension, and the resulting tensor is processed by a convolutional layer to generate the ED sample set X_{ED}.

To alleviate the model's learning burden and encourage the learning of domain-invariant features, we linearly combine the SD sample set X_{SD} and the ED sample set X_{ED} to generate the ID sample set X_{ID}. The formula for generating the ID is as follows:

$$X_{\text{ID}} = \omega X_{\text{SD}} + (1-\omega)\, X_{\text{ED}} \tag{11}$$

where $\omega \in \mathbb{R}^{N\times 1\times 1\times 1}$ is a tensor sampled from a uniform distribution over the interval $(0, 1)$.

B. FAM

To enhance the generalization robustness of feature representations, we construct augmented features in the frequency domain by designing an FAM inspired by the work of Wang et al. [35]. Specifically, the three domain sample sets, X_{SD}, X_{ID}, and X_{ED}, are input to the FAM, which retains the original data X_{SD}, X_{ID}, and X_{ED} while also generating frequency-enhanced representations FX_{SD}, FX_{ID}, and FX_{ED}, collectively forming the multidomain sample set.

Specifically, we first perform a Fourier transform on the input tensor, mapping each channel of the HSI to the frequency domain. Let the input tensor be $X \in \mathbb{R}^{N\times C\times H\times W}$, where N is the batch size, C is the number of channels, and $H \times W$ are the spatial dimensions. A 2-D discrete Fourier transform is applied to each channel of each sample

$$F_{n,c} = F_2\left(X_{n,c}\right) \quad \forall n \in \{1,\ldots,N\}, c \in \{1,\ldots,C\} \tag{12}$$

where, to ensure unbiased feature representation and consistency between the signal's spatial domain energy and frequency-domain energy, $F_2(\cdot)$ represents the 2-D Fourier transform satisfying the orthogonal normalization condition

$$\|F_2(X)\|_2 = \|X\|_2 \tag{13}$$

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the complex spectrum is decomposed into magnitude A and phase Θ

$$A_{n,c} = |F_{n,c}|$$
$$\Theta_{n,c} = \angle F_{n,c} \tag{14}$$

where $|\cdot|$ and $\angle\cdot$ denote the operations of taking the magnitude and phase angle of a complex number, respectively.

After obtaining the magnitude and phase, we add multiplicative noise and additive noise to the magnitude and phase, respectively. Some previous works [36], [37] posit that the phase of an image carries high-level semantic information, and thus, they focus on manipulating the amplitude while keeping the phase unchanged. In contrast, the work of Wang et al. [35] found that perturbing the phase can enhance the generalization ability of the network. Therefore, we add multiplicative noise and additive noise to the magnitude and phase. We define two independent noise fields

$$\alpha_{n,c,h,w} \sim U(a, b)$$
$$\beta_{n,c,h,w} \sim \mathcal{N}\left(\mu_\beta, \sigma_\beta^2\right) \tag{15}$$

where U denotes the uniform distribution, controlling the multiplicative noise intensity, and $\mathcal{N}$ denotes the normal distribution, implementing the additive offset. We apply mixed noise perturbation to both the magnitude and phase

$$A'_{n,c} = \alpha_{n,c} \odot A_{n,c} + \beta_{n,c}$$
$$\Theta'_{n,c} = \alpha_{n,c} \odot \Theta_{n,c} + \beta_{n,c} \tag{16}$$

where $\odot$ denotes the Hadamard product (element-wise multiplication). The complex spectrum is reconstructed from the modified magnitude and phase as

$$F'_{n,c} = A'_{n,c} \odot \exp\left(i\Theta'_{n,c}\right) \tag{17}$$

expanding into real and imaginary parts

$$F'_{n,c} = A'_{n,c} \odot \cos\left(\Theta'_{n,c}\right) + iA'_{n,c} \odot \sin\left(\Theta'_{n,c}\right) \tag{18}$$

the time-domain image is recovered through the orthogonal inverse Fourier transform

$$\hat{X}_{n,c} = \mathrm{Re}\left[F_2^{-1}\left(F'_{n,c}\right)\right] \tag{19}$$

the final output satisfies

$$\hat{X} \in \mathbb{R}^{N\times C\times H\times W}, \mathbb{E}\left[\|\hat{X} - X\|_2\right] > 0 \tag{20}$$

through the aforementioned formulas, we obtain the frequency-enhanced representations of the three domain sample sets, thus constructing the multidomain sample set.

C. Discriminator Based on Self-Knowledge Distillation

To fully leverage the multidomain sample set obtained from the FAM, we propose a discriminator integrated with a self-knowledge distillation mechanism. This discriminator generates classification tensors $C_D \in \mathbb{R}^{N\times K}, D \in \{\mathrm{SD}, \mathrm{ID}, \mathrm{ED}\}$ and $F_D \in \mathbb{R}^{N\times K}, D \in \{\mathrm{SD}, \mathrm{ID}, \mathrm{ED}\}$ for the input multidomain sample set, where N is the batch size and K is the number of classes. The classification tensor, after undergoing the Softmax function, is utilized to compute the cross-entropy loss $\mathcal{L}_{\mathrm{ce}}$. $\mathcal{L}_{\mathrm{ce}}$ can be represented as

$$\mathcal{L}_{\mathrm{ce}}(p_i, y_i) = -\sum_k y_i^k \log p_i^k \tag{21}$$

where y_i represents the one-hot encoding of the label for sample x_i, c denotes the class index, and p_i is the probability distribution obtained by applying the Softmax function to the classification vector. Therefore, the cross-entropy loss for the classification tensor C_{SD} can be defined as

$$\mathcal{L}_{\mathrm{SD}}(C_{\mathrm{SD}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i) \tag{22}$$

where Y_{SD} and N represent the SD labels and the batch size, respectively. To ensure the accuracy of the generated ED samples, the discriminator also needs to classify them correctly. Furthermore, the discriminator also needs to make correct predictions for the ID and the frequency-enhanced representations. Following the consistency regularization proposed in FixMatch [38], we can assume that X_{ID}, X_{ED}, $\mathrm{FX}_{\mathrm{SD}}$, $\mathrm{FX}_{\mathrm{ID}}$, and $\mathrm{FX}_{\mathrm{ED}}$ share the same label space as X_{SD}, because these sample sets are all generated by applying certain transformations to the SD. Therefore, the labels Y_{SD} of the SD are used to calculate the classification loss for X_{ID}, X_{ED}, $\mathrm{FX}_{\mathrm{SD}}$, $\mathrm{FX}_{\mathrm{ID}}$, and $\mathrm{FX}_{\mathrm{ED}}$

$$\mathcal{L}_{\mathrm{ID}}(C_{\mathrm{ID}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i)$$
$$\mathcal{L}_{\mathrm{ED}}(C_{\mathrm{ED}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i)$$
$$\mathcal{L}_{\mathrm{FSD}}(F_{\mathrm{SD}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i)$$
$$\mathcal{L}_{\mathrm{FID}}(F_{\mathrm{ID}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i)$$
$$\mathcal{L}_{\mathrm{FED}}(F_{\mathrm{ED}}, Y_{\mathrm{SD}}) = \frac{1}{N}\sum_i \mathcal{L}_{\mathrm{ce}}(p_i, y_i) \tag{23}$$

therefore, the total classification loss can be written as

$$\mathcal{L}_{\mathrm{cls}} = \sum_{D\in\{\mathrm{SD},\mathrm{ID},\mathrm{ED}\}} (\mathcal{L}_D + \mathcal{L}_{\mathrm{FD}}). \tag{24}$$

To achieve the construction of domain-invariant knowledge from different domains, we propose a sample pair distillation loss, denoted as $\mathcal{L}_{\mathrm{pairKD}}$. Specifically, $\mathcal{L}_{\mathrm{pairKD}}$ pairs data with the same label but originating from different domains, penalizing the overall mismatch of each logit. This measure avoids bias toward a specific domain, thereby encouraging the learning of domain-invariant features. For a set of paired samples (X_i, X_j) with the same semantic label but originating from two different domains within a batch, a class prototype vector is computed by averaging their predicted probability distributions. The KL divergence is then used to constrain the alignment of individual sample distributions with this class prototype. This bi-domain averaging strategy is more suitable for single-SD cross-scene HSI classification. The predicted probability distributions are inevitably affected by intraclass spectral variations. Averaging the predictions of samples from the same class within a batch as the class prototype undoubtedly increases the model's robustness. Furthermore, unlike

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traditional knowledge distillation, $\mathcal{L}_{\text{pairKD}}$ does not require the addition of extra network components. Instead, it operates as a self-knowledge distillation mechanism, learning from pairs of samples from two domains within a batch for model updating.

Let X_D^y and FX_D^y denote two sets of samples with the same class y from two different domains $D \in \{\text{SD}, \text{ID}, \text{ED}\}$ within a batch. We then obtain the overall set of logits by taking the average of the logits for these two different domain sample sets

$$\bar{z}\left(X_D^y \cup \text{FX}_D^y\right) = \sum_{x_i \in X_D^y, \text{FX}_D^y} \frac{2z\left(x_i\right)}{\left|X_D^y + \text{FX}_D^y\right|} \tag{25}$$

where $\bar{z}(X_D^y \cup \text{FX}_D^y)$ represents the average logits for two different sample sets belonging to class y within a batch. Serving as the "class prototype" to construct cross-domain consensus, this average logit $\bar{z}(X_D^y \cup \text{FX}_D^y)$ is subsequently used to calculate the posterior probability distribution. We can then obtain the posterior probability distribution formed by the data X_D^y and FX_D^y based on $\bar{z}(X_D^y \cup \text{FX}_D^y)$ as

$$\bar{P}\left(k \mid X_D^y \cup \text{FX}_D^y; \theta, \tau\right) = \frac{\exp\left(\bar{z}_k\left(X_D^y \cup \text{FX}_D^y\right)/\tau\right)}{\sum_{i=1}^{K} \exp\left(\bar{z}_i\left(X_D^y \cup \text{FX}_D^y\right)/\tau\right)} \tag{26}$$

for conciseness, we denote $X_D^y \cup \text{FX}_D^y$ by $\mathcal{X}_D^y$. therefore, $\mathcal{L}_{\text{pairKD}}$ can be expressed as

$$\mathcal{L}_{\text{pairKD}}\left(\mathcal{X}_D^y; \theta, \tau\right) = \sum_{x_i \in \mathcal{X}_D^y} \sum_{k=1}^{K} D_{\text{KL}}\left(\bar{P}\left(k \mid \mathcal{X}_D^y; \bar{\theta}, \tau\right) \parallel P\left(k \mid x_i; \theta, \tau\right)\right) \tag{27}$$

where $\bar{\theta}$ is a fixed copy of the parameters θ. Ultimately, $\mathcal{L}_{\text{pairKD}}$ promotes cross-domain transferability by averaging the logits of samples from the same class but two different domains to generate a cross-domain consensus class prototype, and then using KL divergence to penalize the prediction distribution of individual samples. Additionally, similar to some methods employing contrastive learning [7], [10], [39], we introduce a supervised contrastive loss $\mathcal{L}_{\text{sup}}$ and a contrastive adversarial loss $\mathcal{L}_{\text{adv}}$. The formula for the supervised contrastive loss is as follows:

$$\mathcal{L}_{\text{sup}} = -\sum_{i=0}^{N} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\left(S\left(f_i, f_p^+\right)/\tau\right)}{\sum_{a \in A(i)} \exp\left(S\left(f_i, f_a^-\right)/\tau\right)} \tag{28}$$

where, for each feature f_i in a mini-batch, $P(i)$ and $A(i)$ are the sets of positive and negative samples, respectively. $|P(i)|$ is the number of positive samples, and f_p^+ and f_a^- are one of the positive and negative samples, respectively. N represents the batch size, and the $S(\cdot)$ function is a measure of similarity between two features, generally employing dot product or cosine distance. τ is the temperature coefficient. $\mathcal{L}_{\text{sup}}$ is used to optimize the discriminator. During the optimization of the discriminator, features belonging to the same class in SD, ID, and ED are placed in $P(i)$, and features from different classes are placed in $A(i)$. $\mathcal{L}_{\text{sup}}$ optimizes the discriminator to make samples of the same class closer and samples of different classes farther apart.

Considering the effectiveness of the generator in producing the ED, we introduce a contrastive adversarial loss $\mathcal{L}_{\text{adv}}$. The formula for $\mathcal{L}_{\text{adv}}$ is as follows:

$$\mathcal{L}_{\text{adv}} = -\sum_{k} \sum_{i=0}^{n_k} \frac{1}{|P_k(i)|} \sum_{p \in P_k(i)} \log \frac{\exp\left(S\left(f_i, f_p^+\right)/\tau\right)}{\sum_{a \in A_k(i)} \exp\left(S\left(f_i, f_a^-\right)/\tau\right)} \tag{29}$$

where n_k is the number of samples in the kth class, $P_k(i)$ and $A_k(i)$ are the positive and negative sample sets for the kth class, respectively. Samples belonging to the kth class in SD belong to the positive sample set, and ED or ID is randomly selected at each iteration to construct the negative sample set. During the optimization of the generator, samples of the kth class in SD are separated from samples of the kth class in ED or ID. Through adversarial training, the ED generated by the generator contains domain-specific information, thereby prompting the discriminator to learn from a more diversified data distribution.

D. Training Phase

In DPSKDnet, the generator and the discriminator are optimized separately. First, with the multidomain sample set as input, the discriminator is optimized with the following objective:

$$\min_{\theta_D} \mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{sup}} + \lambda_2 \mathcal{L}_{\text{pairKD}} \tag{30}$$

where λ_1 and λ_2 are hyperparameters used to balance the supervised contrastive loss and the sample pair distillation loss. Furthermore, to optimize the generator, in addition to the contrastive adversarial loss$\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{ED}}$ is used as an auxiliary loss to ensure that the ED resides in the correct label space. The generator's optimization objective is as follows:

$$\min_{\theta_G} \mathcal{L} = \mathcal{L}_{\text{ED}} + \mathcal{L}_{\text{adv}}. \tag{31}$$

IV. Experiment

This section primarily evaluates and discusses the performance of DPSKDnet on three publicly available cross-scene HSI datasets: HyRANK, Houston, and Pavia. Furthermore, we analyze the improvement effects of different modules and explore the model parameters.

A. Datasets

HyRANK Dataset: The HyRANK dataset comprises two scenes: Dioni and Loukia. The images for these scenes were acquired by the Hyperion sensor (EO-1 and USGS). The Dioni scene image has spatial dimensions of 250 × 1376 pixels, while the Loukia scene image has dimensions of 249 × 945 pixels. Both scenes contain 176 identical spectral bands and share the same 12 classes. Detailed information regarding the image classes and their respective sample counts is provided in Table I. The pseudo-color images and ground-truth maps are depicted in Fig. 3.

Houston Dataset: The Houston Dataset incorporates data from two scenes, Houston2013 [40] and Houston2018 [41].

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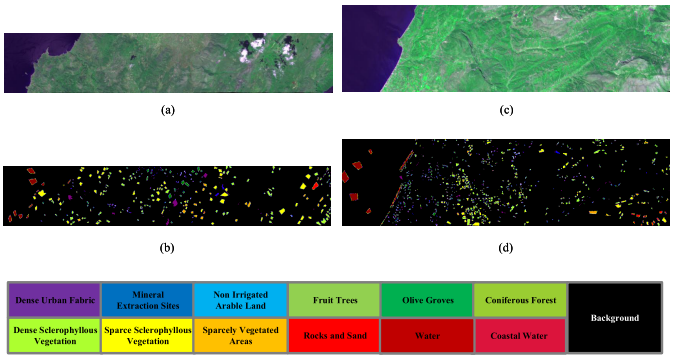


Fig. 3. Pseudo-color images and ground-truth maps of the HyRANK dataset. (a) Pseudo-color image of Dioni. (b) Ground-truth map of Dioni. (c) Pseudo-color image of Loukia. (d) Ground-truth map of Loukia.

TABLE I
HyRANK Dataset: Land-Cover Types and Sample Counts

No.	Land cover type	Dioni (SD)	Loukia (TD)
1	Dense Urban Fabric	1262	288
2	Mineral Extraction Sites	204	67
3	Non Irrigated Arable Land	614	542
4	Fruit Trees	150	79
5	Olive Groves	1768	1401
6	Coniferous Forest	361	500
7	Dense Sclerophyllous Vegetation	5035	3793
8	Sparse Sclerophyllous Vegetation	6374	2803
9	Sparsely Vegetated Areas	1754	404
10	Rocks and Sand	492	487
11	Water	1612	1393
12	Coastal Water	398	451
	Total	20024	12208

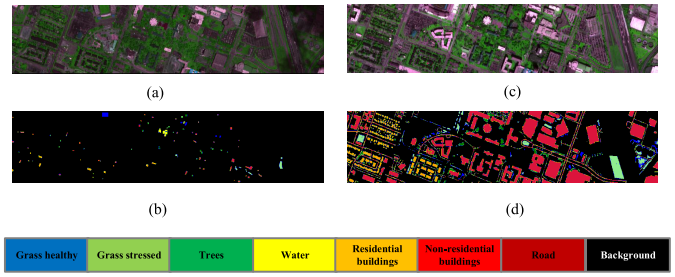


Fig. 4. Pseudo-color images and ground-truth maps of the Houston dataset. (a) Pseudo-color image of Houston2013. (b) Ground-truth map of Houston2013. (c) Pseudo-color image of Houston2018. (d) Ground-truth map of Houston2018.

These were captured by dissimilar sensors at different times over the University of Houston campus and its surrounding areas. The Houston2013 dataset comprises 349 × 1095 pixels with 144 spectral bands, covering a wavelength range of 380–1050 nm and exhibiting a spatial resolution of 2.5 m. In contrast, the Houston2018 dataset, while sharing the same wavelength range, contains 48 spectral bands and offers a finer spatial resolution of 1 m. Seven consistent classes are present across these two scenes. For the experiments, we extracted the 48 spectral bands from the Houston2013 scene that correspond to the spectral bands in the Houston2018 scene, spanning the wavelength range of 0.38–1.05 μm. A common overlapping region of 209×955 pixels was selected

TABLE II
Houston2013 and Houston2018 Datasets: Land-Cover Types and Sample Counts

No.	Land cover type	Houston2013 (SD)	Houston2018 (TD)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

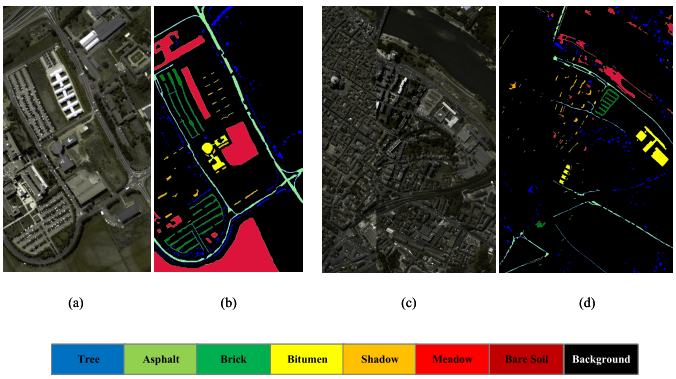


Fig. 5. Pseudo-color images and ground-truth maps of the Pavia dataset. (a) Pseudo-color image of UP. (b) Ground-truth map of UP. (c) Pseudo-color image of PC. (d) Ground-truth map of PC.

TABLE III
Pavia Dataset: Land-Cover Types and Sample Counts

No.	Land cover type	Pavia University (SD)	Pavia Center (TD)
1	Trees	3064	7598
2	Asphalt	6634	9248
3	Bricks	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
	Total	39332	39355

for analysis. Table II details the classes and their respective sample counts. Furthermore, the pseudo-color images and ground-truth maps of the dataset are illustrated in Fig. 4.

Pavia Dataset: The Pavia dataset includes two scenes—Pavia University (UP) and Pavia Center (PC)—both acquired using the reflective optics system imaging spectrometer (ROSIS) sensor. These scenes share a common spectral range spanning 430–860 nm. The UP scene is characterized by 103 spectral bands, dimensions of 610 × 340 pixels, and a spatial resolution of 1.3 m. In contrast, the PC scene comprises 1096 × 715 pixels and 102 spectral bands. The last spectral band of UP was removed to equalize the number of spectral bands with PC. Both scenes share the same seven land-cover classes, with the names and sample counts of these classes detailed in Table III. Fig. 5 illustrates their respective pseudo-color images and ground-truth maps.

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TABLE IV
CLASSIFICATION PERFORMANCE ON THREE DATASETS UNDER DIFFERENT LRS

LR	OA			KC		
	HyRANK	Houston	Pavia	HyRANK	Houston	Pavia
1e-2	61.07	68.92	82.72	51.20	52.60	79.23
1e-3	**62.75**	**80.04**	**83.24**	**53.56**	**65.49**	**79.81**
1e-4	60.30	73.69	82.99	50.06	52.67	79.61
1e-5	52.78	49.51	57.28	37.51	26.75	49.17

B. Implementation Details

Regarding the implementation details, the proposed DPSKDnet was implemented using the PyTorch framework and trained on a single NVIDIA RTX 3090 GPU. Following the fundamental setup of DG methods, only the source scene was accessible during the training phase, and the trained model was applied to the target scene during testing. On the Houston dataset, we selected 80% of the samples for training, and the SD was augmented by a factor of four using random flipping and random spectral noise. For the Pavia and HyRANK datasets, 50% of the samples were chosen for training. Concerning the specific parameters of DPSKDnet, the training epoch was set to 400, the patch size p to 13, the batch size to 256, the learning rate (LR) to 1e-3, and the Adam optimizer was employed. For the parameters of the FAM, the multiplicative noise was sampled using a uniform distribution, with the window set to $[0.8, 1.1]$. The additive noise was sampled using a normal distribution, with the mean set to 0.1 and the standard deviation set to 0.05. The parameter analysis regarding the window settings and noise distributions can be found in Section IV-C. The parameters for the perturbation in the generator are initialized with a standard normal distribution (mean of 0, variance of 1). To compare the classification performance, we adopted three evaluation metrics [42]: overall accuracy (OA), per-class accuracy, and the Kappa coefficient (KC).

C. Parameter Analysis

This section evaluates the robustness of the proposed DPSKDnet across the three TDs through a systematic parameter sensitivity analysis. The experiments focus on examining several key hyperparameters, including the model LR, the discriminator projection dimension, the regularization parameters λ_1 and λ_2, and the parameter range for the multiplicative noise and the distribution of the additive noise in FAM. Specifically, the model LR was explored within the range of {1e-2, 1e-3, 1e-4, 1e-5}. The discriminator projection dimension was investigated using the values {32, 64, 128, 256, 512}. For the regularization parameters λ_1 and λ_2, the values considered were {1e+ 1, 1e+ 0, 1e-1, 1e-2, 1e-3}. For the parameter range of the multiplicative noise, experiments were conducted within $[0.2, 0.5]$, $[0.5, 0.8]$, $[0.8, 1.1]$, and $[1.1, 1.4]$. The distribution for the additive noise was chosen between the normal distribution and the uniform distribution.

Table IV presents the classification results across the three datasets for various base LRs. Across all three datasets, a consistent optimal base LR of 1e-3 was identified. Under other LRs, the OA and KC metrics for all three datasets consistently declined, with a more pronounced downward trend observed when the LR was less than 1e-3. We infer that this may be due to the dynamic perturbation destylization generating overly complex sample features, leading to insufficient model updates at lower LRs and, consequently, performance degradation.

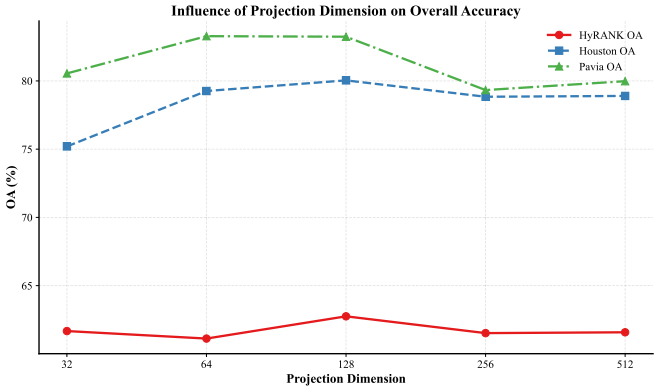


Fig. 6. Effect of different projection dimensions on OA across the three datasets.

The projection dimension of the discriminator constitutes a critical hyperparameter in the generative adversarial network. A higher projection dimension enables the discriminator to learn more complex features and input representations. However, this can also lead to the discriminator becoming excessively powerful prematurely, thereby hindering the generator's training and its ability to effectively extract domain-invariant features. Conversely, a lower projection dimension limits the discriminator's capacity to model complex data patterns. While this can mitigate overfitting on small datasets, it may also impede the discriminator from effectively guiding the generator to produce high-quality samples. Fig. 6 illustrates the trends in OA on the HyRANK, Houston, and Pavia hyperspectral datasets when employing different projection dimensions for the discriminator. The experimental results indicate that a projection dimension of 128 for the discriminator yields optimal performance on the HyRANK and Houston datasets. Although a slight decrease in model performance is observed on the Pavia dataset at this dimension, the magnitude of this reduction is negligible. Consequently, we will fix the projection dimension to 128 in the subsequent experiments.

λ_1 and λ_2 are crucial hyperparameters for DPSKDnet, as they govern the contribution of the supervised contrastive loss and the sample pair distillation loss to DG. When λ_1 is set too large, the model enforces an excessively strict separation between intraclass and interclass samples from the SD, ID, and ED domains. This may cause the model to overemphasize interclass separability while sacrificing other important objectives, such as domain alignment or the quality of samples generated by the generator. In this case, the features of intraclass samples may become overly compact, converging

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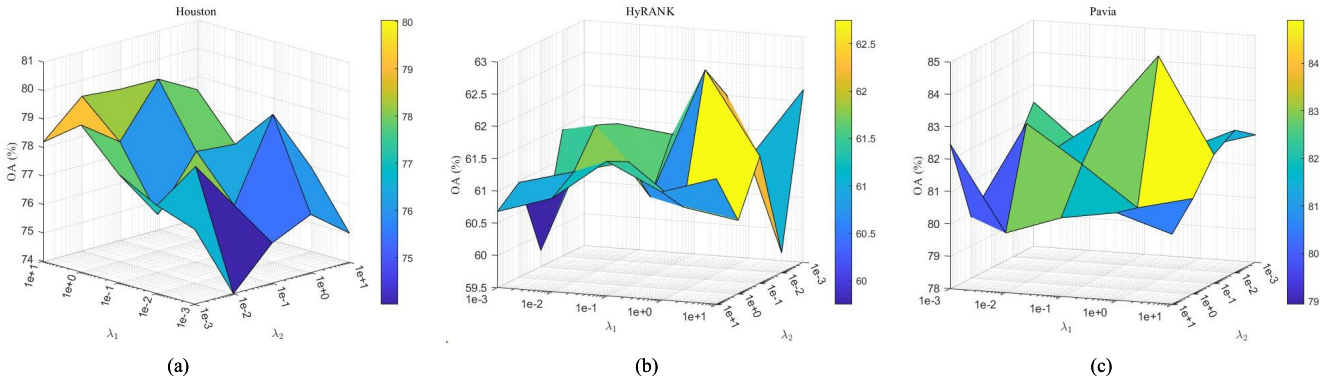


Fig. 7. OA of DPSKDnet with different λ_1 and λ_2. (a) Houston dataset. (b) HyRank dataset. (c) Pavia dataset.

TABLE V
PERFORMANCE (OA) ON THREE DATASETS UNDER DIFFERENT HYPERPARAMETER DISTRIBUTION SETTINGS

Uniform Distribution			
Parameter Range	**HyRANK**	**Houston**	**Pavia**
[0.2, 0.5]	61.20	74.50	78.90
[0.5, 0.8]	60.10	73.19	78.27
[0.8, 1.1]	61.91	71.17	78.59
[1.1, 1.4]	60.59	75.13	78.48
Normal Distribution			
Parameter Range	**HyRANK**	**Houston**	**Pavia**
[0.2, 0.5]	61.37	76.53	79.93
[0.5, 0.8]	60.64	75.89	77.15
[0.8, 1.1]	**62.32**	**79.60**	**84.91**
[1.1, 1.4]	60.34	74.71	78.27

toward a single point and thereby reducing the diversity of feature representations.

When λ_2 is excessively large, the model tends to overemphasize output consistency, making the regularization too stringent and weakening the discriminability between different classes. Furthermore, the loss term associated with λ_2 is a soft-label constraint; if the gradient becomes too dominant, it can suppress the hard-label signal from the $\mathcal{L}_{\text{cls}}$, ultimately leading to a decline in classification accuracy.

Fig. 7 illustrates the trends in classification accuracy of DPSKDnet across the three experimental datasets for different combinations of λ_1 and λ_2. As depicted in Fig. 7, the optimal parameters for λ_1 and λ_2 are 1e+ 0 and 1e-1, respectively, across all three datasets. Consequently, we will maintain the parameters for λ_1 and λ_2 at 1e+ 0 and 1e-1 for all three datasets in the subsequent experiments.

The parameter range of the multiplicative noise and the distribution of the additive noise have a significant impact on the performance of FAM. As shown in Table V, when the parameter range of the multiplicative noise is set to $[0.8, 1.1]$ and the additive noise follows a normal distribution, FAM achieves the best performance across all three datasets. Therefore, we retain this parameter setting for the subsequent experiments.

D. Comparison With Advanced Cross-Scene Classification Methods

To comprehensively evaluate the performance of DPSKDnet, we implemented a series of state-of-the-art cross-scene classification methods under the same experimental platform. These include DG methods for RGB images such as SagNets [43] and DSU [44], knowledge distillation-based DG method XDED [32], data augmentation-based method FACT [37], as well as a set of advanced cross-scene HSI classification methods, including SDEnet [7], LLURnet [10], LDGnet [8], and EHSnet [18].

To ensure the reliability of the experimental results, each method was independently repeated five times using different random seeds, and the results are reported as mean $\pm$ standard deviation. The specific parameter settings for each method are as follows. For the Houston dataset, 80% of the samples were selected for training, and the SD was augmented fourfold through random flipping and random spectral noise. For the Pavia and HyRANK datasets, 50% of the samples were used for training. The patch size p was set to 13 for all methods. All other parameters strictly follow those in the original papers. The number of training epochs was set to 100 for FACT, XDED, SagNets, and DSU; 200 for LDGnet and EHSnet; and 400 for SDEnet and LLURnet. The batch size was set to 64 for FACT, SagNets, DSU, and LLURnet, and 256 for XDED, LDGnet, SDEnet, and EHSnet. The LR was set to 1e-3 for FACT, SagNets, DSU, XDED, SDEnet, and LLURnet, and 1e-2 for LDGnet and EHSnet. All methods were optimized using the Adam optimizer.

Tables VI–VIII showcase the performance of different methods on the cross-scene HSI classification task across the three datasets. In conventional DG approaches, the main categories include RGB-based methods, pure data augmentation techniques, and knowledge distillation frameworks. Specifically, the XDED model achieves the highest OA and KC on the Houston 2018 and PC datasets, while the FACT model obtains the best OA and KC on the Loukia dataset. In contrast, the proposed DPSKDnet demonstrates significantly superior performance in terms of both OA and KC compared with these traditional DG methods (including RGB-based, data augmentation, and knowledge distillation approaches), highlighting its strong cross-DG capability.

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TABLE VI
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE LOUKIA DATASET

Class	FACT [37]	SagNets [43]	DSU [44]	XDED [32]	SDEnet [7]	LLURnet [10]	LDGnet [8]	EHSnet [18]	DPSKDnet
1	32.56	20.58	21.65	20.39	24.66	30.97	**42.43**	13.01	31.07
2	**18.61**	15.19	3.70	10.74	12.22	2.96	5.55	16.67	7.04
3	31.24	**71.45**	52.63	50.66	46.62	51.27	45.87	39.06	43.57
4	0.00	**29.37**	21.52	5.82	14.18	7.60	22.79	21.26	17.22
5	**32.63**	0.02	0.11	1.37	1.21	0.27	0.00	7.30	0.96
6	3.92	14.98	27.77	16.68	32.37	33.27	**42.32**	32.75	32.32
7	74.92	76.58	79.45	**83.64**	76.37	73.95	77.39	78.50	80.25
8	64.13	66.39	65.66	54.06	**71.91**	68.70	48.14	54.83	71.50
9	45.01	20.60	23.16	19.50	47.42	26.01	**73.83**	59.55	33.88
10	0.44	0.00	0.31	10.60	0.00	**27.82**	0.80	20.53	0.00
11	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.99	**100.0**	**100.0**	**100.0**
12	**100.0**	**100.0**	99.95	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
OA	61.48±0.67	60.09±0.84	60.53±1.36	58.89±1.85	62.09±0.82	61.20±1.35	58.64±1.44	60.38±1.46	**62.57±0.17**
KC	52.17±1.27	50.21±1.04	50.65±1.45	48.38±2.12	52.72±0.86	51.89±1.48	49.38±1.52	50.96±1.91	**53.07±0.32**

TABLE VII
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE HOUSTON2018 DATASET

Class	FACT [37]	SagNets [43]	DSU [44]	XDED [32]	SDEnet [7]	LLURnet [10]	LDGnet [8]	EHSnet [18]	DPSKDnet
1	21.02	20.31	30.86	32.64	42.45	43.40	**50.66**	48.44	35.79
2	65.38	37.70	49.32	70.25	82.19	80.34	80.43	78.97	**84.10**
3	43.40	38.19	38.46	53.59	**64.64**	59.14	54.76	60.01	53.77
4	78.82	80.91	92.73	92.73	**100.0**	**100.0**	89.09	85.45	**100.0**
5	66.05	64.26	**76.36**	64.24	70.46	65.19	69.08	66.64	54.35
6	94.65	**97.09**	92.02	92.76	88.79	56.84	90.79	85.24	90.19
7	12.04	1.26	33.28	23.56	41.83	**62.56**	31.93	55.69	59.66
OA	74.66±1.09	71.84±2.13	75.15±2.40	75.98±1.19	78.29±0.75	76.34±1.19	77.72±1.65	77.02±2.17	**79.10±0.57**
KC	51.34±2.18	41.05±6.34	54.68±4.26	55.44±3.32	62.65±1.45	60.77±2.07	60.72±3.67	61.96±2.51	**63.96±0.77**

TABLE VIII
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE PC DATASET

Class	FACT [37]	SagNets [43]	DSU [44]	XDED [32]	SDEnet [7]	LLURnet [10]	LDGnet [8]	EHSnet [18]	DPSKDnet
1	89.37	93.74	71.74	92.51	85.67	85.93	**94.88**	92.70	90.42
2	84.40	80.63	78.72	**84.70**	83.79	81.36	82.48	83.75	84.45
3	**73.88**	35.91	17.12	29.81	69.22	67.35	48.07	51.36	61.81
4	16.79	21.74	55.56	67.07	77.93	81.34	72.95	74.01	**85.78**
5	85.43	88.32	79.50	**91.18**	86.03	90.83	88.13	85.99	85.30
6	76.21	67.71	62.13	71.91	**78.48**	75.60	66.16	72.89	74.28
7	78.40	74.75	**86.84**	77.89	76.28	78.70	72.27	80.60	78.42
OA	70.55±1.38	67.77±2.23	68.99±3.66	77.52±1.57	80.56±1.33	81.06±1.16	78.18±1.60	80.21±2.58	**82.56±0.43**
KC	64.99±1.49	61.17±2.82	62.58±4.42	72.82±1.99	76.67±1.61	77.26±1.35	73.74±1.89	76.15±3.13	**78.97±0.51**

More notably, when compared with existing HSI-specific DG methods, DPSKDnet consistently outperforms all competitors across all three benchmark datasets in both OA and KC, confirming its robustness and superior adaptability to diverse hyperspectral scenes. Furthermore, hyperspectral datasets often suffer from class imbalance in both source and TDs, with this issue being particularly severe in the Loukia dataset. Such an imbalance frequently causes many methods (e.g., FACT and SagNets) to converge to locally optimal solutions for dominant classes, thereby failing to achieve globally optimal generalization performance. In this context, incorporating semantic information has been regarded as a potentially effective auxiliary strategy for mitigating class imbalance. However, the success of these methods largely depends on the quality and reliability of the introduced semantic information. For instance, LDGnet and EHSnet exhibit inferior performance on the Loukia dataset, likely due to ineffective or inaccurate semantic guidance. In particular, LDGnet may be misled by incorrect semantic cues, causing the model to be trapped in class-specific optima and fail to reach a globally optimal generalization.

Overall, due to the inherent class imbalance in hyperspectral data, different methods exhibit varying levels of performance and bias toward specific categories. Importantly, DPSKDnet achieves the highest OA and KC values and the lowest standard deviations across all datasets, conclusively demonstrating its robustness, consistency, and superiority in cross-scene HSI classification.

Furthermore, beyond the quantitative analysis conducted on the three cross-scene HSI classification datasets, Figs. 8–10

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TABLE IX
TRAINING TIME (IN SECONDS) OF ONE EPOCH IN DIFFERENT METHODS ON THREE TASKS

Task	FACT [37]	SagNets [43]	DSU [44]	XDED [32]	SDEnet [7]	LLURnet [10]	LDGnet [8]	EHSnet [18]	DPSKDnet
HyRANK	148.96	**2.51**	46.22	2.74	16.19	16.99	40.47	35.34	4.58
Houston	10.50	**0.71**	8.68	1.43	4.55	6.62	18.86	14.05	2.34
Pavia	96.10	**2.90**	55.30	2.98	20.51	24.39	68.07	54.24	6.19

TABLE X
ABLATION STUDY ON THE OA AND KC METRICS ACROSS THREE DATASETS (BEST RESULTS IN BOLD BLACK)

Variants	Modules			OA			KC		
	G	$\mathcal{L}_{\text{pairKD}}$	FAM	Houston	HyRANK	Pavia	Houston	HyRANK	Pavia
Source only (Baselines)	×	×	×	74.81	60.49	77.43	59.47	51.19	72.87
w/o G	×	✓	✓	77.98	61.47	80.91	63.05	51.94	77.28
w/o $\mathcal{L}_{\text{pairKD}}$	✓	×	✓	79.90	62.12	82.17	65.34	52.53	78.59
Only G	✓	×	×	76.68	61.93	78.85	61.71	53.05	74.73
Only FAM	×	×	✓	77.15	61.56	81.11	60.37	51.93	77.29
DPSKDnet	✓	✓	✓	**80.04**	**62.75**	**83.24**	**65.49**	**53.56**	**79.81**

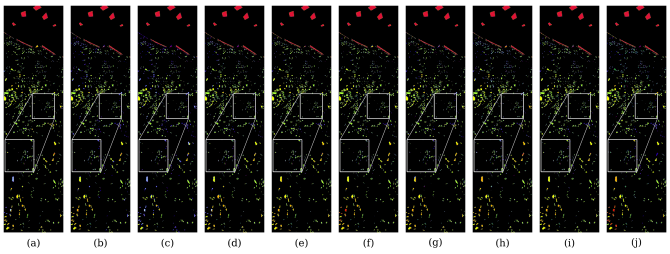


Fig. 8. Visualized classification maps obtained by different methods on the HyRANK dataset. (a) FACT (61.95%). (b) SagNets (60.09%). (c) DSU (60.45%). (d) XDED (59.59%). (e) SDEnet (61.85%). (f) LLURnet (61.67%). (g) LDGnet (60.01%). (h) EHSnet (60.60%). (i) DPSKDnet (62.75%). (j) Ground truth.

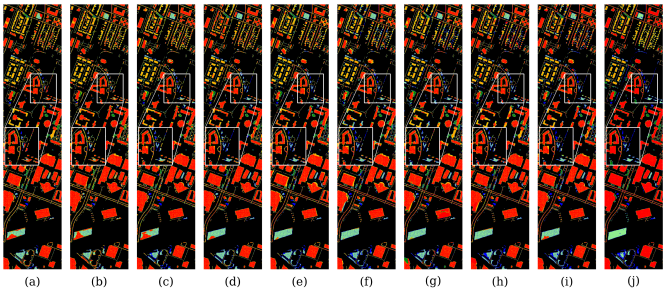


Fig. 9. Visualized classification maps obtained by different methods on the Houston dataset. (a) FACT (74.42%). (b) SagNets (73.27%). (c) DSU (74.81%). (d) XDED (76.69%). (e) SDEnet (78.61%). (f) LLURnet (77.43%). (g) LDGnet (77.73%). (h) EHSnet (77.82%). (i) DPSKDnet (80.04%). (j) Ground truth.

present the visualized classification maps of different methods. As can be observed from these figures, the classification map generated by DPSKDnet exhibits a closer resemblance to the ground-truth land-cover distribution, with clearer boundaries and fewer misclassifications.

In addition, we evaluated the computational complexity of different methods on the three datasets. Table IX lists the training time per epoch for all experimental data. The reported execution time per epoch was obtained by averaging the time

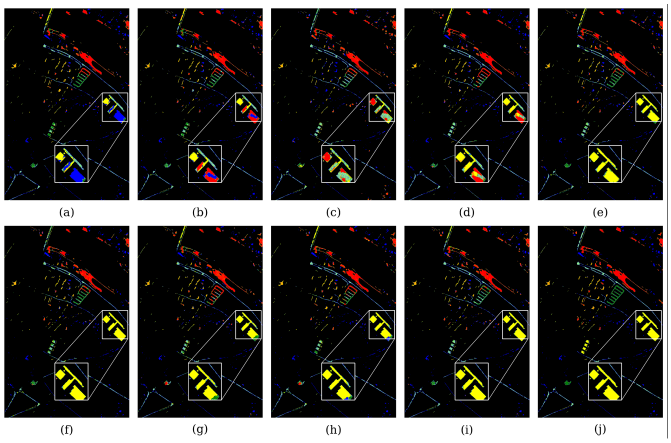


Fig. 10. Visualized classification maps obtained by different methods on the Pavia dataset. (a) FACT (70.16%). (b) SagNets (69.35%). (c) DSU (68.31%). (d) XDED (77.73%). (e) SDEnet (80.54%). (f) LLURnet (81.30%). (g) LDGnet (78.63%). (h) EHSnet (80.40%). (i) DPSKDnet (83.24%). (j) Ground truth.

taken to train each model for 100 epochs. As shown in Table IX, SagNets demonstrates the best training efficiency across the three datasets, benefiting from its simple structure. XDED performs second best. DPSKDnet is in third place but still outperforms the HSI-based DG methods. SDEnet and LLURnet exhibit relatively stable, mid-range training times across the three datasets. Due to the integration of language information, EHSnet and LDGnet have relatively longer training times across all three datasets. FACT incurs longer training times when the training set is large due to its data augmentation operations.

E. Ablation Study

1) Overall Ablation Study: In this section, ablation experiments are primarily conducted to validate the contribution of each component within DPSKDnet. "G" denotes the generator. Specifically, Table X presents the ablation experimental results under different metrics. Table X lists all the different versions

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TABLE XI
ABLATION STUDY OF THE LOSS FUNCTION ON THE OA AND KC METRICS ACROSS THREE DATASETS (BEST RESULTS IN BOLD BLACK)

Variants	Modules				OA			KC		
	$\mathcal{L}_{cls}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{sup}$	$\mathcal{L}_{pairKD}$	Houston	HyRANK	Pavia	Houston	HyRANK	Pavia
only $\mathcal{L}_{cls}$	✓	×	×	×	72.39	59.75	74.71	56.77	50.36	69.75
add $\mathcal{L}_{adv}$	✓	✓	×	×	74.15	61.64	81.41	58.65	51.94	77.63
add $\mathcal{L}_{sup}$	✓	✓	✓	×	77.16	61.36	81.91	61.74	51.51	78.29
DPSKDnet (add $\mathcal{L}_{pairKD}$)	✓	✓	✓	✓	**79.60**	**62.32**	**84.91**	**64.90**	**52.70**	**81.85**

of the model, where each model is designed by removing one of the components while keeping the others consistent.

It is noteworthy that the first variant, trained solely on the SD without any domain extension, serves as the baseline model. A significant performance gap exists between the baseline model and DPSKDnet (for the Houston, HyRANK, and Pavia datasets: OA difference: 5.23%, 2.26%, 5.81%; KC difference: 6.02%, 2.37%, 6.94%). This indicates that the proposed DG algorithm effectively enhances the model's out-of-distribution performance. As anticipated, DPSKDnet achieves the best performance when all components are integrated, while removing any component results in varying degrees of performance degradation. Furthermore, the removal of the generator section precludes the generation of ED. Consequently, in the variant model without the proposed generator, we adopted the generator from SDEnet as a replacement to validate the effectiveness of the generator based on dynamic perturbation destylization. As evidenced by the "w/o G" row in Table X, a certain degree of decline in both OA and KC is observed after replacing the dynamic perturbation generator with the SDEnet generator. This demonstrates that the dynamic perturbation-based destylization generation paradigm employed by the generator is a more effective solution for generating ED.

Regarding the ablation of $\mathcal{L}_{pairKD}$, a certain degree of decline in both OA and KC is observed upon its removal compared to the complete model. However, the performance with $\mathcal{L}_{pairKD}$ ablated still demonstrates a significant improvement in both OA and KC compared to the baseline model. This suggests that the combination of the generator and the FAM has a positive contribution to performance. Nevertheless, the model achieves its optimal results when $\mathcal{L}_{pairKD}$ is integrated.

Concerning the model variant retaining only the generator, its performance surpasses that of the baseline model, indicating a certain contribution of the generator based on dynamic perturbation destylization to the generalization performance. Similarly, the variant retaining only the FAM also outperforms the baseline, suggesting a positive impact of constructing enhanced representations in the frequency domain on generalization. The complete DPSKDnet achieves the highest OA and KC values across the three datasets, significantly outperforming the other variants. This demonstrates that the generator based on dynamic perturbation destylization, $\mathcal{L}_{pairKD}$, and the FAM mutually promote each other, collectively enhancing the overall classification accuracy and classification consistency of the model.

2) Ablation Study of the Loss Function: To further validate the effectiveness of each component within the loss function, we conducted ablation experiments by adding individual loss components while keeping the others unchanged. Table XI presents the ablation experimental results under different metrics.

Table XI presents the results of the loss function ablation experiments conducted on the Houston, HyRANK, and Pavia datasets. The experimental results clearly demonstrate that the model performance improves progressively as each loss component is added, confirming the effective contribution of every loss term to the final performance.

To begin with, we use only the classification loss $\mathcal{L}_{cls}$ as the baseline. When the adversarial loss $\mathcal{L}_{adv}$ is introduced into the generator, the model can generate samples with better class discriminability, leading to a noticeable performance improvement—particularly on the Pavia dataset, where both OA and KC increase significantly. Next, after incorporating the supervised contrastive loss $\mathcal{L}_{sup}$ into the discriminator, the model performance continues to improve, especially on the Houston dataset. It is worth noting, however, that a slight performance drop occurs on the HyRANK dataset after adding $\mathcal{L}_{sup}$. This may be attributed to the inherent complexity of the HyRANK dataset, suggesting that $\mathcal{L}_{sup}$ alone may not be sufficient for the discriminator to capture its essential class characteristics. Finally, by introducing the sample pair distillation loss $\mathcal{L}_{pairKD}$, the model achieves its best performance, yielding consistent and comprehensive improvements across all datasets.

3) Effectiveness Analysis of the Generator Architecture: In this study, the generator performs morphological operations only on grayscale images, and the dynamic perturbation destylization module is not incorporated into the morphological branch. This section analyzes the rationale behind this design choice.

In this study, the design choice of applying morphological operations only to grayscale images is primarily based on the inherent characteristics of morphological processing. Traditional morphological operations (such as erosion and dilation) are typically performed based on the intensity or geometric information within pixel neighborhoods, aiming to separate or connect fundamental structural elements such as edges, skeletons, or textures. These operations are highly sensitive to absolute pixel intensity values rather than color vectors.

Directly applying morphological operations to each RGB channel independently may lead to interchannel imbalance and distortion, introducing undesirable color artifacts or biases that can compromise the overall image style and structural integrity. Converting the color image to grayscale effectively separates structural information from color (or style)

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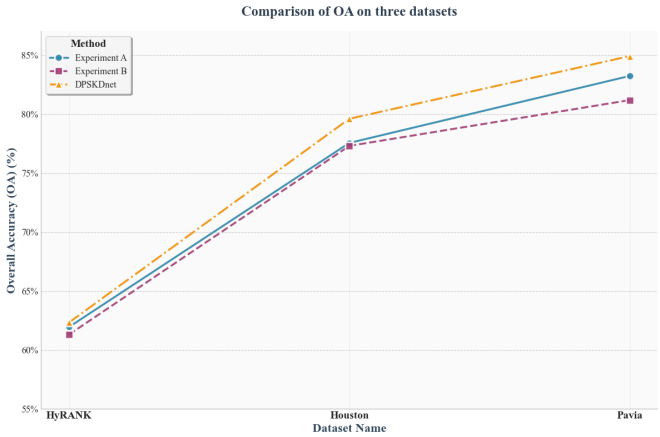


Fig. 11. OA metric comparison of Experiment A (morphological operations on all three RGB channels) and Experiment B (dynamic perturbation destylization module in the morphological branch) on all three datasets.

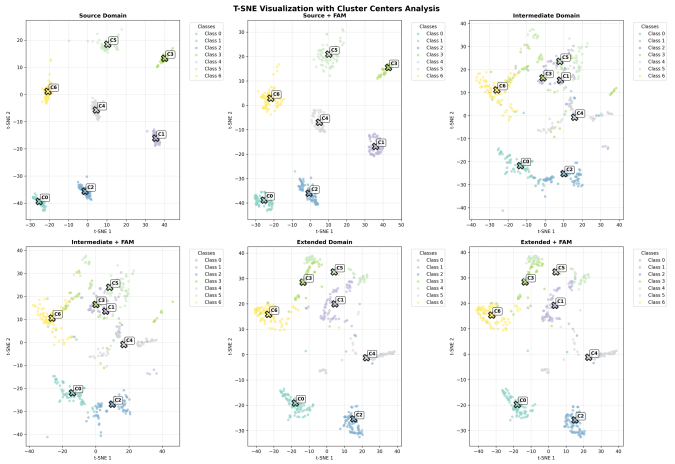


Fig. 12. t-SNE comparative visualization of class-wise cluster centers of logits from various domains (SD, ID, and ED) and the new domains generated by FAM.

information. By performing morphological operations on grayscale images, the model can focus on extracting geometric and shape-related structures, enabling the morphological branch to capture the essential geometric features of the target in a more pure and style-independent manner.

The reason for not employing the destylization module on the morphological branch is that the primary goal of this branch is to extract fundamental, stable geometric structures. During the destylization process, there is a risk of inadvertently damaging or smoothing out important structural information (such as fine edges or connections) that the morphological branch is trying to preserve, thereby weakening its effectiveness.

To validate this design choice, we included supplementary verification experiments.

1) *Experiment A:* Morphological operations are applied to all three RGB channels.
2) *Experiment B:* The dynamic perturbation destylization module is forcibly included in the morphological branch.

The experimental results are shown in Fig. 11. As shown in Fig. 11, both Experiments A and B show a decrease in the OA metric on all three datasets compared to DPSKDnet, thus validating the effectiveness of the generator design.

4) Effectiveness Analysis of $\mathcal{L}_{pairKD}$: $\mathcal{L}_{\text{pairKD}}$ constructs a cross-domain shared class prototype by averaging the logits of samples belonging to the same category but originating from different domains. The reliability of this averaging operation is based on a key assumption: the distributional discrepancy of logits for the same class across domains should be relatively small. If the domain shift is too significant, the simple averaging operation may produce a blurred or distorted prototype, thereby reducing the effectiveness of knowledge distillation.

In this study, $\mathcal{L}_{\text{pairKD}}$ is applied to paired samples between a given domain and its augmented counterpart generated by the FAM, aiming to enhance cross-domain knowledge consistency. To evaluate the effectiveness of this loss, we first employ the trained generator to produce the ED, followed by noise augmentation on the source, intermediate, and EDs using FAM. The trained discriminator is then used to predict logits for all input samples. We visualize these logits using t-SNE and extract the cluster centers of each class to compare the differences among domains. If the cluster centers of the same class across different domains exhibit minor discrepancies, it indicates that $\mathcal{L}_{\text{pairKD}}$ effectively promotes consistency in cross-domain class representations.

As shown in Fig. 12, the cluster centers of each domain and their corresponding FAM-generated domains exhibit slight shifts, indicating that the domain shift is not severe. This observation suggests that the logits preserve sufficient cross-domain semantic consistency, enabling their mean values to effectively represent the core semantic information of each class, thereby forming a reliable and domain-shared knowledge prototype.

V. Conclusion

In this study, we propose DPSKDnet, a DG framework for cross-scene HSI classification. By introducing a generator based on dynamic perturbation destylization and a sample pair self-distillation strategy, the model effectively learns domain-invariant representations without requiring any data from the TD. Experimental results on three public HSI datasets demonstrate that DPSKDnet achieved state-of-the-art results across all three TDs: Loukia, Houston2018, and PC. Specifically, the OA improved by 0.48%, 0.81%, and 1.5%, respectively, and the KC improved by 0.35%, 1.31%, and 1.71%, respectively. Despite these promising results, the framework still faces limitations in low-resource scenarios, where excessive synthetic samples may introduce distributional bias. Future work will focus on developing adaptive augmentation mechanisms and conducting broader evaluations across real-world HSI scenes.

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